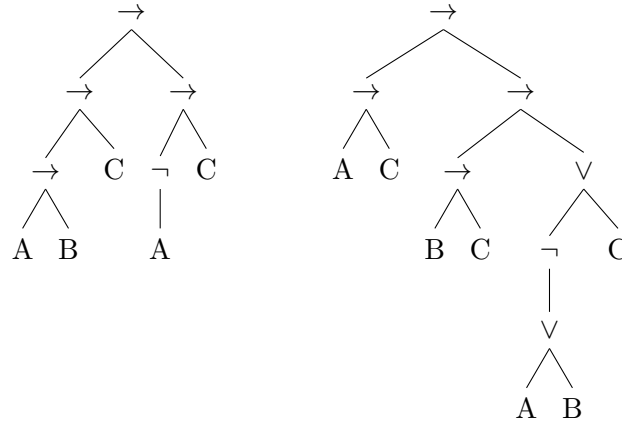


Exercise Sheet 12f - Solutions

Propositional Logic – Natural Deduction

1. The parse trees of F and G are



2. Here is a constructive Natural Deduction proof of $((A \rightarrow B) \rightarrow C) \rightarrow \neg A \rightarrow C$

$$\begin{array}{c}
 \frac{\overline{A}^3 \quad \overline{\neg A}^2}{\perp} [\neg E] \\
 \frac{\perp}{B} [\perp E] \\
 \frac{(A \rightarrow B) \rightarrow C^1 \quad \overline{A \rightarrow B}^3}{A \rightarrow B} [\rightarrow I] \\
 \frac{A \rightarrow B \quad C}{C} [\rightarrow E] \\
 \frac{C}{\neg A \rightarrow C}^2 [\rightarrow I] \\
 \frac{\neg A \rightarrow C}{((A \rightarrow B) \rightarrow C) \rightarrow \neg A \rightarrow C}^1 [\rightarrow I]
 \end{array}$$

3. Here is a classical Natural Deduction proof of $(A \rightarrow C) \rightarrow (B \rightarrow C) \rightarrow \neg(A \vee B) \vee C$

$$\begin{array}{c}
 \frac{\overline{A \rightarrow C}^1 \quad \overline{A}^6}{C} [\rightarrow E] \quad \frac{\overline{B \rightarrow C}^2 \quad \overline{B}^7}{C} [\rightarrow E] \quad \frac{\overline{C}^4}{\neg C} [\neg E] \\
 \frac{A \vee B^5 \quad \frac{\perp}{A \rightarrow \perp}^6 [\rightarrow I] \quad \frac{\perp}{B \rightarrow \perp}^7 [\rightarrow I]}{A \rightarrow \perp \quad B \rightarrow \perp} [\vee E] \\
 \frac{\overline{C}^3}{\neg(A \vee B) \vee C} [\vee I_R] \quad \frac{\perp}{\neg(A \vee B)}^5 [\neg I] \quad \frac{\neg(A \vee B)}{\neg(A \vee B) \vee C} [\vee I_L] \\
 \frac{\overline{C \vee \neg C} [\text{LEM}] \quad \frac{\overline{C \rightarrow \neg(A \vee B) \vee C}^3}{\neg(A \vee B) \vee C} [\rightarrow I] \quad \frac{\neg(A \vee B) \vee C}{\neg C \rightarrow \neg(A \vee B) \vee C}^4 [\rightarrow I]}{\neg(A \vee B) \vee C} [\vee E] \\
 \frac{\neg(A \vee B) \vee C}{(B \rightarrow C) \rightarrow \neg(A \vee B) \vee C}^2 [\rightarrow I] \\
 \frac{(B \rightarrow C) \rightarrow \neg(A \vee B) \vee C}{(A \rightarrow C) \rightarrow (B \rightarrow C) \rightarrow \neg(A \vee B) \vee C}^1 [\rightarrow I]
 \end{array}$$

4. F and G are equivalent as they are both true and therefore are both equivalent to $\top$.

Another explanation is that because both formulas are provable, by consistency they are both valid, which means that they are both true for all valuations, i.e., that they are equivalent.